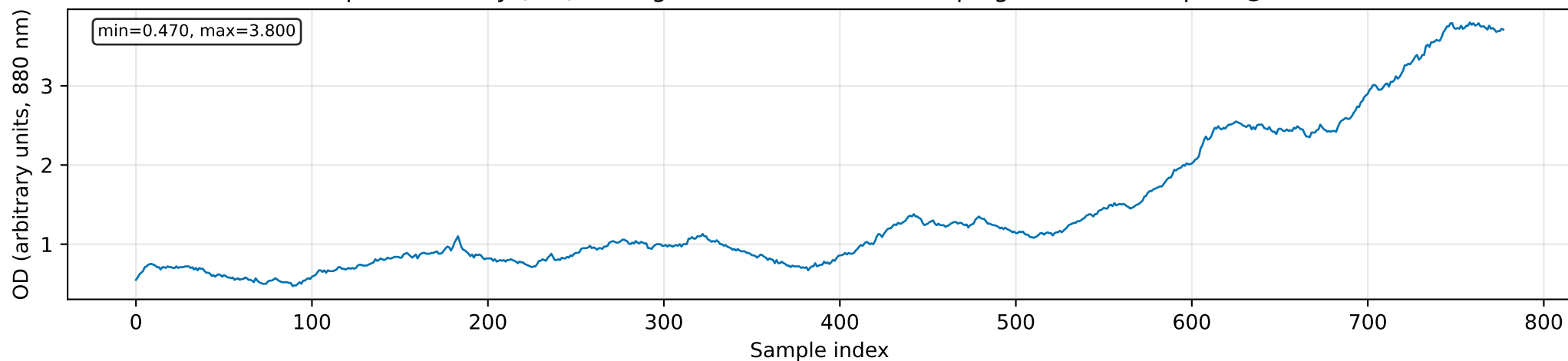
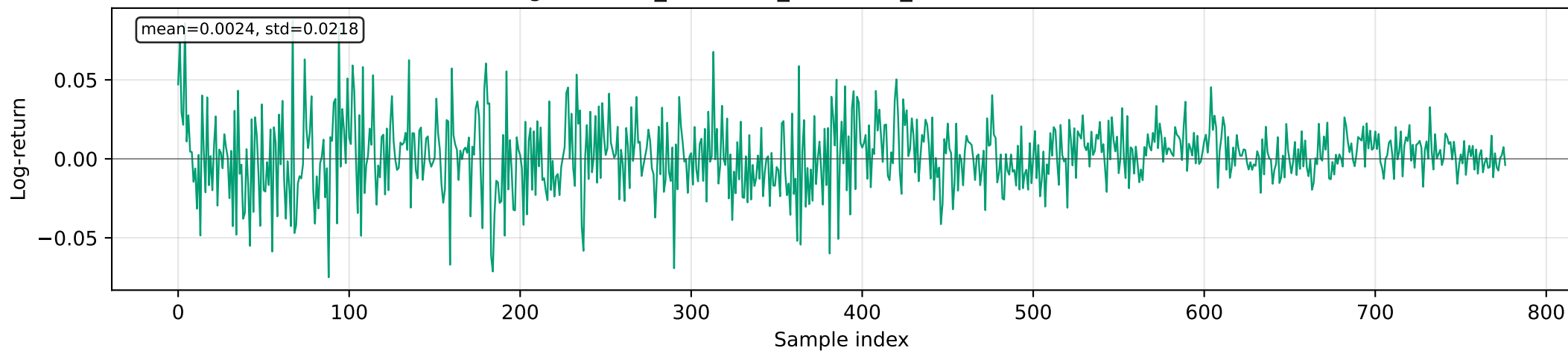


Preprocessing pipeline (Pipeline B) — four stages from raw OD to model-ready input

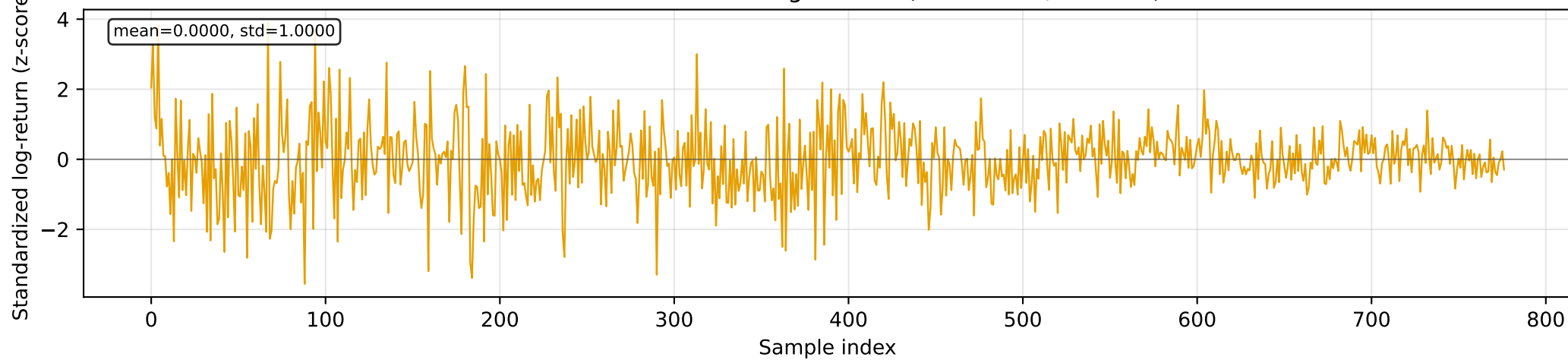
Panel 1: Raw optical density (OD) — single LUCY cultivation campaign, n=778 samples @ 10-min intervals



Panel 2: Log-returns $r_t = \ln OD_t - \ln OD_{(t-1)}$ — first-difference, n=777



Panel 3: Standardized log-returns (zero mean, unit std)



Panel 4: Rescaled to $[-1, 1]$ using global $r_{\min}=-3.559$, $r_{\max}=3.901$ — model training input (Pipeline B)

